



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

INDEX
NUMBER

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BIOLOGY
9648/03

25th August 2016
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]

Answer all questions

Section B]

Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	52
1 [13]	
2 [13]	
3 [14]	
4 [12]	
Section B	20
5a [6]	
5b [6]	
5c [8]	
TOTAL	72

Section A

Answer **all** questions in this section.

- 1 Human Growth Hormone is important to augment normal growth and development in the treatment of individuals with dwarfism. Fig. 1.1 shows how human growth hormone can be produced via expression of recombinant DNA in *Escherichia coli* host cells.

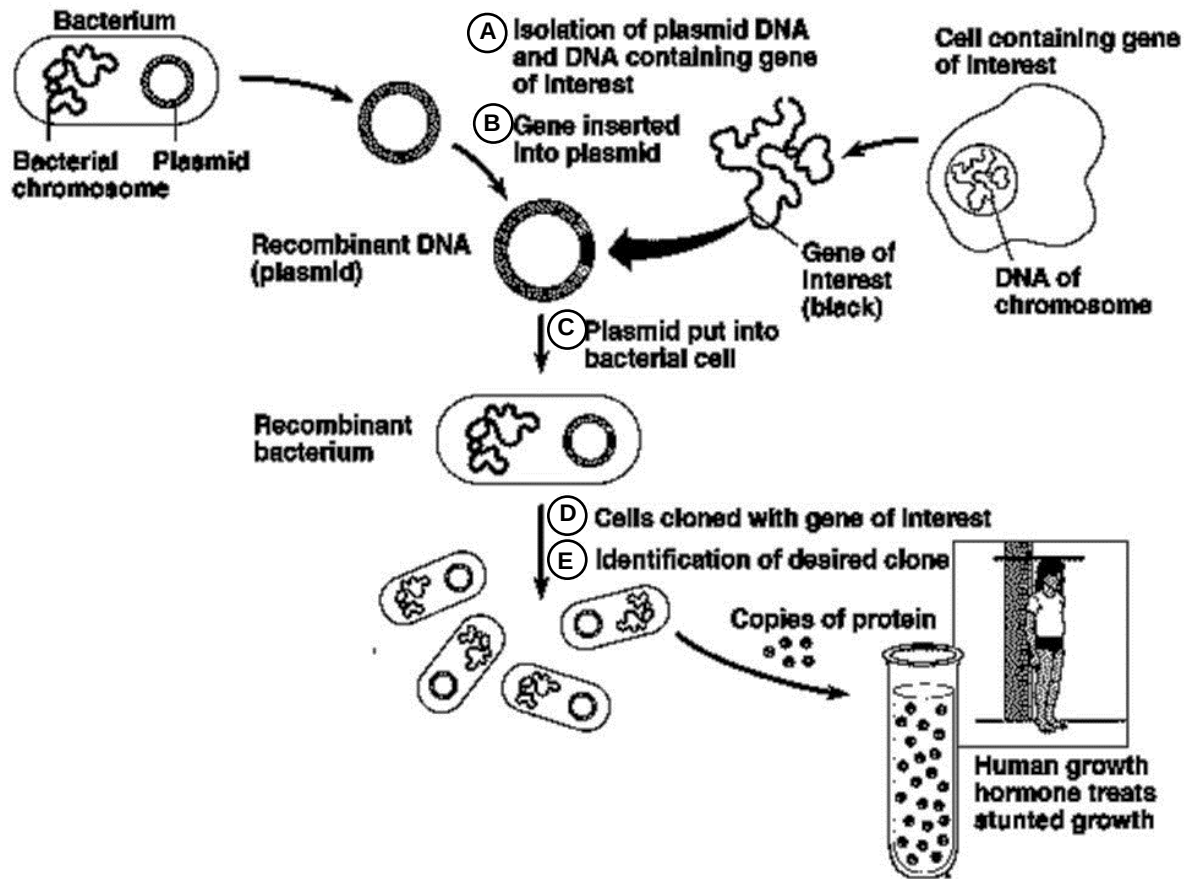


Fig. 1.1

- (a) Explain what is meant by recombinant DNA.

.....
[1]

- (b) Name the process required in the following procedure in Fig 1.1.

C :

E :

[1]

(c) The gene of interest cannot be taken directly from DNA of chromosome but require additional processing in order to produce functional protein.

(i) With reference to Fig. 1.1, explain why the gene of interest cannot be taken directly from chromosomal DNA.

.....
.....
.....[2]

(ii) Outline the additional processing required to yield the gene of interest prior to insertion into the plasmid.

.....
.....
.....
.....[3]

(d) State one possible pair of gene markers present on the cloning site of the plasmid for the identification of desired clone in Fig. 1.1.

.....
.....
.....[1]

(e) Outline the process for Stage E in Fig 1.1 using one of the gene markers in (d).

.....
.....
.....
.....[3]

Fig 1.2 shows details of how stage A and B are carried out.

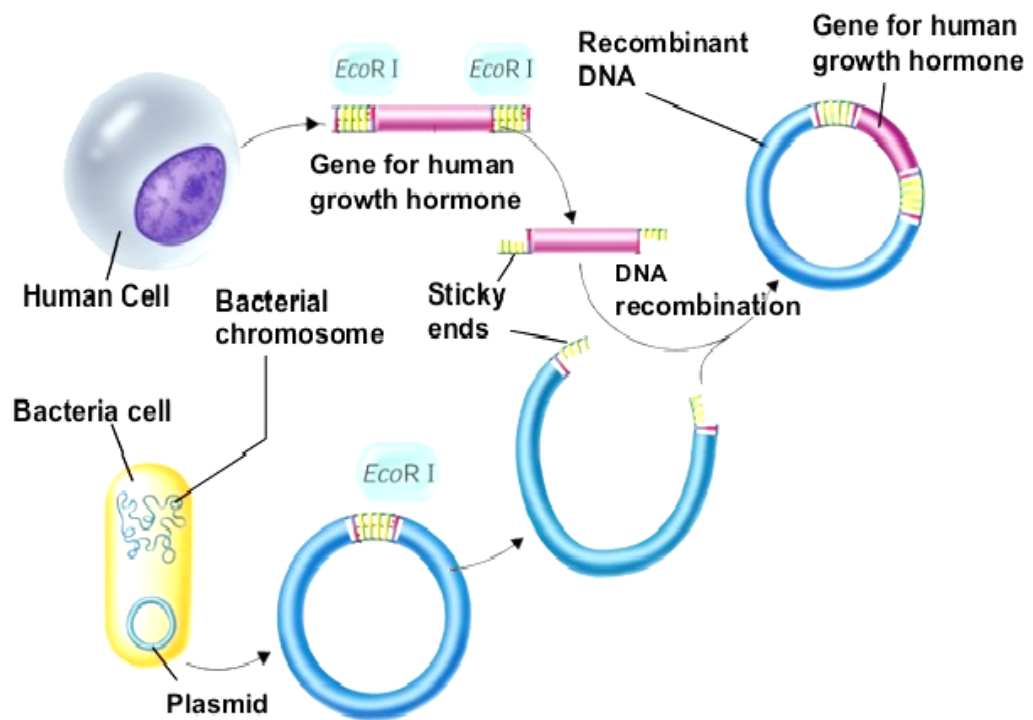


Fig 1.2

- (f) A scientist commented that two different restriction enzymes should be used to isolate the gene for human growth hormone instead of using only *EcoRI* restriction enzyme. Explain the rationale behind his comment.

.....

.....

.....[2]

[Total: 13]

2 Restriction Fragment Length Polymorphism (RFLP) is an important application for comparative genetic analysis of normal and diseased individuals.

(a) Outline the key techniques used for RFLP analysis.

.....

.....

.....

.....

.....[4]

Fig. 2.1 shows the pedigree and southern blots of an RFLP locus that is closely linked with a disease phenotype. Affected individuals are shaded in the pedigree.

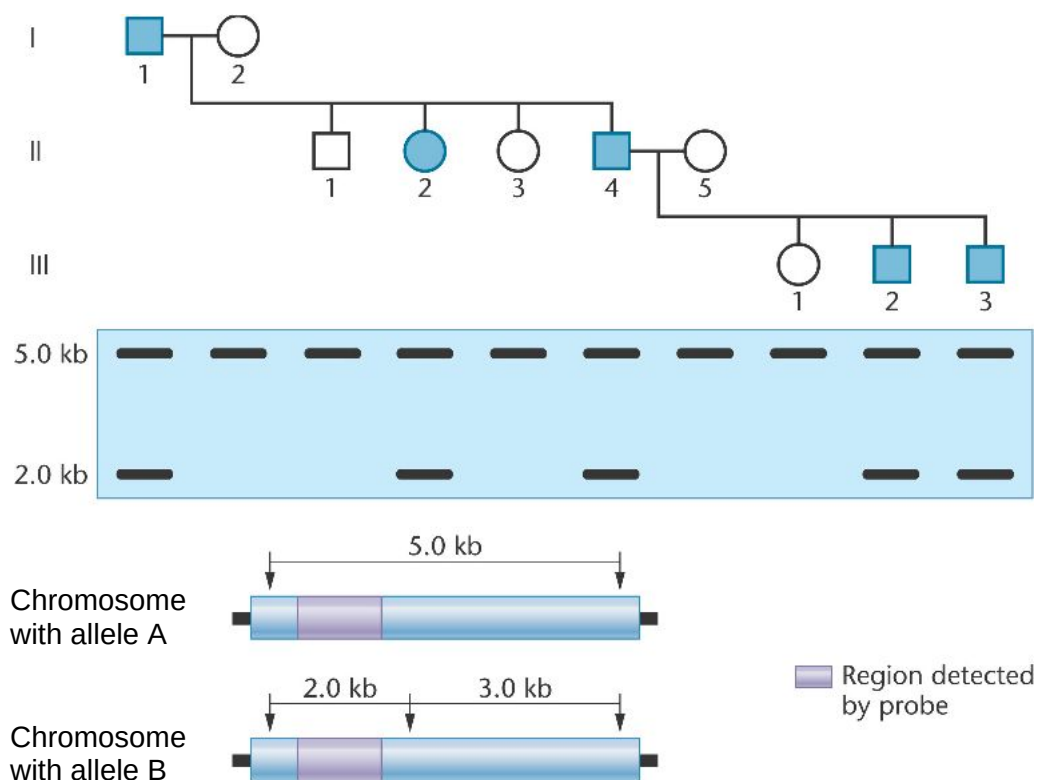


Fig. 2.1

(b) Explain the genetic basis of RFLP in comparative analysis in disease study.

.....

.....

..... [2]

(c) With reference to Fig. 2.1,

(i) Identify the allele responsible for the disease.

.....[1]

(ii) State the considerations for target region of the probe.

.....
.....
.....[2]

(iii) Explain if the disease allele is dominant or recessive.

.....
.....
.....[2]

(d) Briefly describe two other applications of RFLP.

.....
.....
.....[2]

[Total: 13]

3 Fig. 3.1 shows how gene therapy using modified viruses can be carried out.

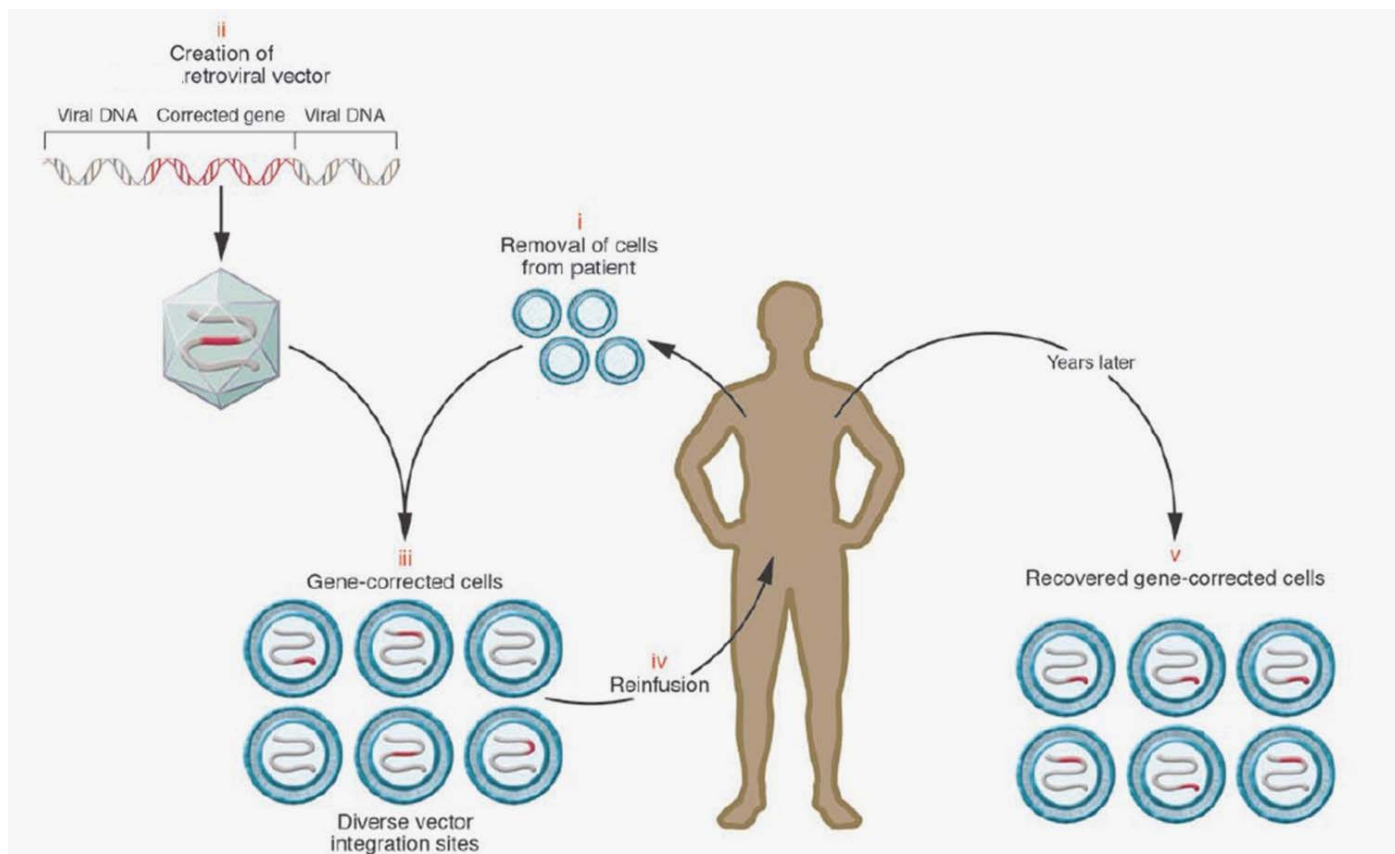


Figure obtained from Retroviral integration and human gene therapy The Journal of Clinical Investigation, Volume 117, Number 8

Fig. 3.1

Severe combined immunodeficiency (SCID) and X-linked SCID are two genetic diseases that can be treated by the gene therapy method shown in Fig. 3.1.

(a) State the genes that can be corrected for SCID and X-linked SCID.

SCID:

X-linked SCID:

[2]

(b) Explain how mutation in one of the genes cited in (a) would give rise to immunodeficiency.

.....

.....

.....[2]

Excerpt from the *The Journal of Clinical Investigation* review titled *Retroviral integration and human gene therapy*:

"However, with these successes came the first serious adverse events in retrovirus based gene therapy. Three of the SCID-X1 patients treated by the French team developed a leukemia-like lymphoproliferative disease."

- (c)** Suggest why these patients developed leukemia-like lymphoproliferative diseases.

.....
.....
.....[2]

In 1999, 18-year old Jesse Gelsinger died after suffering from a massive immune response following a clinical trial that administered adenoviral-based gene therapy.

- (d)** Besides adenovirus' ability to elicit a strong immune response, explain why adenovirus is also not favoured in the treatment of the genetic diseases cited in **(a)**.

.....
.....
.....[2]

- (e)** Explain why scientists prefer to isolate hematopoietic stem cells rather than T-cells from the patient for the gene therapy treatment shown in Fig. 3.1.

.....
.....
.....
.....[3]

Viruses are not the only vectors that are utilised to deliver genes into patients' host cells.

- (f)** State one other type of vector that is not viral.

.....
.....[1]

- (g)** Discuss two ethical implications of gene therapy.

.....
.....
.....
.....[2]

[Total: 14]

4 Planning question

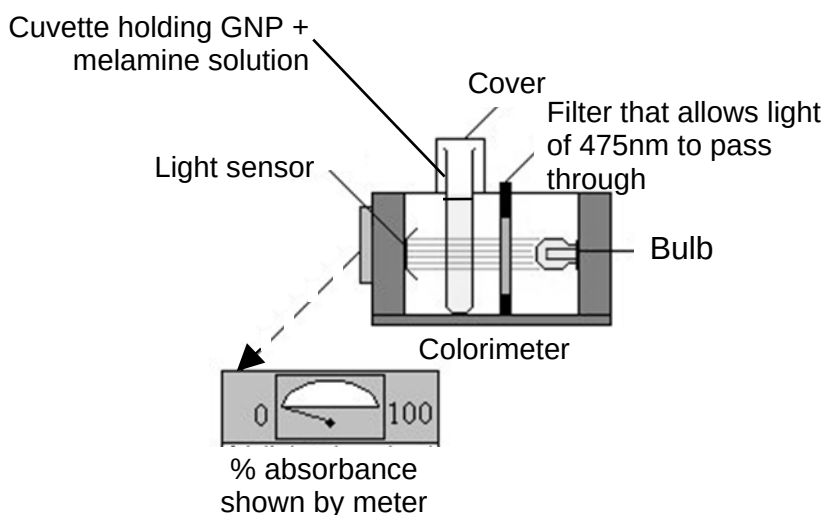
In 2008, contaminated milk and infant formula with melamine in China resulted in over a thousand babies hospitalised and caused several deaths. Melamine with its large number of nitrogen atoms, was unethically added to foods to mimic proteins detected through conventional measurements for nitrogen.

A research group found a way to detect the presence of melamine via a colour indicator. They attached cyanuric acid to gold nanoparticles to produce functionalized gold nanoparticles (GNP) that are red, the cyanuric acid component of GNP detects the melamine through hydrogen-bonding. Upon binding to melamine, the GNP changes from red to blue, giving a clear signal that there is contamination.



The intensity of the blue colour after a set time interval of 5 minutes is a measure of the concentration of the melamine present in a specimen.

The test involves addition of 1 cm³ of solution (to be tested) to 0.1 cm³ of GNP in a cuvette (glass container), followed by colourimetric measurement of absorbance (% absorbance) at a specified wavelength. It is known that the blue wavelength is 475 nm.



You are required to plan, but not carry out, an investigation to determine the lowest concentration of melamine detectable by GNP indicator.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- 1 % GNP
- Colourimeter
- 6 x Cuvettes (container for colourimeter measurement)
- 6 x Test-tubes
- Test-tube rack
- 2 x 10 cm³ syringes
- 2 x 1 cm³ syringes
- Micropipette
- 15 cm³ of 1% melamine solution labelled M
- 50 cm³ of buffer solution
- Marker pen
- Stopwatch
- Safety goggles

- Disposable gloves

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- proposed layout of results tables with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

[illegible]

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This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.

Section B: Free-response question

Answer **all** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5**
- (a)** Compare between Genomic and cDNA libraries used in protein production. [6]
 - (b)** Explain the advantages and limitations of PCR. [6]
 - (c)** Discuss the pros and cons of genetically-modified crop plants. [8]

[Total: 20]

END OF PAPER

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